

# 1 Notation

We model four events each year: growth, colonization, fizzle and burnout.

Let  $(k, t)$  be the time index, where  $k$  represent the year index and  $t$  represent the season index, where  $t = 0$  for the time at the beginning of the year,  $t = 1$  for the time after growth,  $t = 2$  for the time after colonization,  $t = 3$  for the time after fizzle and  $t = 4$  for the time after burnout. So  $(k, 0)$  represents the same time as  $(k - 1, 4)$

Assume there are  $n$  patches and we label them with  $p_i$ . For each patch we use  $(\mathbb{1}_i^{(k,t)}, N_i^{(k,t)})$  to describe its state.  $N_i^{(k,t)}$  represents population size in patch  $p_i$  at time point  $(k, t)$ , and  $\mathbb{1}_i^{(k,t)}$  is defined as:

$$\mathbb{1}_i^{(k,t)} = \begin{cases} 1, & \text{if } p_i \text{ is infected at time } (k, t), \\ 0, & \text{otherwise} \end{cases}$$

Let  $U \sim \text{Uniform}(0, 1)$  be a random variable uniformly distributed over  $(0, 1)$ , and we have

$$\mathbb{1}_{U \leq P} = \begin{cases} 1, & \text{if } U \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

## 2 Growth

We can assume that reproduction of rats follows logistic growth, i.e.,

$$N_i^{(k,1)} = N_i^{(k,0)} + rN_i^{(k,0)}\left(1 - \frac{N_i^{(k,0)}}{K}\right)$$

where  $r$  is the growth rate and  $K$  is the carry capacity of a patch. We currently assume that all patches are of equal area and have the same carrying capacity.

## 3 Colonization

### 3.1 Population size

We assume that each individual rat move with probability  $c$  and land in a random patch. We ignore different distances and assume that all patches are of equal availability to a moving rat. So

$$N_i^{(k,2)} = (1 - c)N_i^{(k,1)} + \frac{c \sum_i N_i^{(k,1)}}{n}$$

The first term is population staying in the patch and the second term is population moving into the patch.

### 3.2 Infection

When an infected host dies the fleas are released and can travel between patches (such as being carried by a moving individual) and infect hosts in other patches.

Let  $z$  be the final size. The number of individuals that died of the disease in the previous year in a patch is  $zDN_i^{(k-1,3)}$  if patch  $p_i$  is infected, and 0 if not. So the total number of death is  $\sum_i zDN_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}$ .

Assume the probability of a susceptible patch getting infected is proportional to the number of death in the previous year, i.e.,

$$P_1^{(k)} = \frac{\alpha z D \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}}{nK}$$

$\alpha \in [0, 1]$  is a scaling constant and  $nK$  is used to ensure  $P_I^{(k)} \leq 1$

And

$$\mathbb{1}_i^{(k,2)} = 1 - (1 - \mathbb{1}_i^{(k,1)}) \mathbb{1}_{U < P_1^{(k)}}$$

## 4 Fizzle

Fizzle probability is

$$P_f = \frac{1}{\mathcal{R}_0}$$

And

$$\mathbb{1}_i^{(k,3)} = \mathbb{1}_i^{(k,2)} \mathbb{1}_{U < P_f}$$

Population size does not change.

## 5 Burnout

In infected patches, the population should be reduced:

$$N_i^{(k,4)} = (1 - zD \mathbb{1}_i^{(k,3)}) N_i^{(k,3)}$$

Burnout probability if a patch didn't fizzle is

$$P_b = q^{Ny^*}$$

( $q$  and  $y^*$  are defined in burnout paper)

And

$$\mathbb{1}_i^{(k,4)} = \mathbb{1}_i^{(k,3)} \mathbb{1}_{U < P_b}$$